

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 313 — Past Questions & Materials
by Renance

Course: CHM 313
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — CHM 313 Instrumental Methods of Analysis

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE

COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

FIRST SEMESTER EXAMINATION, 2024/2025 SESSION APRIL 2025

COURSE TITLE: INSTRUMENTAL METHODS OF ANALYSIS

COURS CODE: CHM313 COURSE UNIT: 3

INSTRUCTION: ANSWER THREE QUESTIONS ONLY TIME: 2HOURS

QUESTION ONE

(a) Explain the principle behind electrochem

(b) i. List 5 advantages of electrochemical m

techniques 5marks

ii. What is the significance of equivalent po

(c) How does potentiometric titration differ from traditional titration methods? 5marks

QUESTION TWO

(a) (i) Explain briefly the principle behind Differential Thermal Analysis (DTA)

ical methods of analysis. 10marks

ethods of analysis over other analytical

10marks

(ii) What information can you obtain from a

(b) Distinguish between Differential Scanning C

Thermal Analysis (DTA). 5marks

(c) List 5 applications of DSC in materia

DTA curve? 5marks

Calorimetry (DSC) and Differential

Thermal Analysis (DTA) and pharmaceutical sciences. 5marks

QUESTION THREE

(a) List three each of optical methods of analysis and electro analytical methods of analysis. _ :

6marks

(b) With the aid of a diagram, show the electromagnetic spectrum with the wavelength in cm

10marks

(c) State the three associated components of a Mass Spectrometer and their functions

marks.

QUESTION FOUR

of absorption in ultraviolet-visible

(a) Discuss the phenomenon of electromagnetic radiation

spectroscopy. 10marks

(b) Derive the Beer Lambert L

10marks

(c) A $6 \times 10^{-5} \text{ M}$ solution of a compound. Z has an absorbance of 0.850 at 700nm wavelength in a 1.0cm cell. what is the molar absorptivity of the solution? 5marks

four marks FIVE

(a) Briefly explain the phenomenon of Turbidity. 5marks

(b) Differentiate between Turbidimetry and Nephelometry 6marks

(c) i. How is Turbidimetry or Nephelometry measured? 8marks ii. What is the basic unit of Turbidity? 3marks.

Law of a reflected, absorbed and transmitted light in a medium.

END OF PAPER

Paper 2 of 2 — CHM 313 INSTRUMENTAL METHODS OF ANALYSIS 2024/2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE

COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

FIRST SEMESTER EXAMINATION, 2024/2025 SESSION APRIL 2025

COURSE TITLE: INSTRUMENTAL METHODS OF ANALYSIS

COURS CODE: CHM313. COURSE UNIT: 3

INSTRUCTION: ANSWER THREE QUESTIONS ONLY TIME: 2HOURS

QUESTION ONE

- (a) Explain the principle behind electrochemical methods of analysis. 10marks
- (b) i. List 5 advantages of electrochemical methods of analysis over other analytical techniques 5marks
- ii. What is the significance of equivalent point in potentiometric titration? 5marks
- (c) How does potentiometric titration differ from traditional titration methods? 5marks

QUESTION TWO

- (a) (i) Explain briefly the principle behind Differential Thermal Analysis (DTA) 10marks :
- (ii) What information can you obtain from a DTA curve? 5marks
- (b) Distinguish between Differential Scanning Calorimetry (DSC) and Differential Thermal Analysis (DTA). 5marks
- (c) List 5 applications of DSC in material and pharmaceutical sciences. 5marks

QUESTION THREE

- (2) List three each of optical methods of analysis and electro analytical methods of analysis. 6marks
- (b) With the aid of a diagram, show the electromagnetic spectrum with the wavelength in cm. 10marks
- (c) State the three associated components of a Mass Spectrometer and their functions. 10marks.

QUESTION FOUR

- (a) Discuss the phenomenon of electromagnetic radiation absorption in ultraviolet-visible spectroscopy. 10marks
- (b) Derive the Beer Lambert Law of a reflected, absorbed and transmitted light in a medium. 10marks
- (c) A $6 \times 10^{-7} \text{M}$ solution of a compound, Z has an absorbance of 0.850 at 700nm wavelength in a 1.0cm cell. what is the molar absorptivity of the solution? 5marks

QUESTION FIVE

- (a) Briefly explain the phenomenon of Turbidity. 8marks

(b) Differentiate between Turbidimetry and Nephelometry 6marks YN.

(c) i. How is Turbidimetry or Nephelometry measured? 3marks ii, What is the basic unit at Turbidity'? 3marks.

END OF PAPER